

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed biodegradable material formulations for porous electrodes, integrating flexible and transparent features sourced from natural polymers for direct laser writing fabrication.
- Designed and implemented methodologies for universal fabrication of **Laser Induced Graphene (LIG)** on biopolymers, achieving electrode materials suitable for electrochemical sensing applications.
- Integrated fabricated electrodes with green sensing active layers to enable the detection of biomarkers within e-textile platforms.
- Collaborated with cross-departmental colleagues and external project partners to develop novel integrated sensing e-textile platforms with reduced environmental footprints.
- Executed Python scripts to automate data extraction and processing for material characterisation, reducing manual effort by approximately 40% for the research team.
- Managed production incidents for enterprise applications using **Splunk** and **Dynatrace**, contributing to the resolution of P1/P2 issues within defined SLA targets.
- Containerised Python-based data processing tools using **Docker** and deployed them to **AWS EC2** instances for scalable execution of material analysis tasks.

SKILLS

Materials Science – Biodegradable materials, Laser Induced Graphene (LIG), Porous electrodes, Direct Laser Writing (DLW), Nanomaterials, Micro-scale fabrication

Chemistry – Electrochemistry, Green chemistry, Material formulation, Biomarker detection

Data Analysis & Programming – Python, pandas, NumPy, SQL, R, RStudio, Scikit-learn

Visualization & BI – Power BI, Tableau, DAX, Power Query

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Software Engineering – Java, Spring Boot, REST APIs, Hibernate, JPA, JUnit, Microservices

PROJECTS

Development of Biodegradable LIG Electrodes for E-Textile Sensors

- Engineered novel biodegradable polymer blends designed for direct laser writing of porous LIG structures, focusing on material stability and conductivity.
- Developed and validated fabrication processes for LIG electrodes on flexible biopolymers, achieving consistent surface morphology and electrochemical performance.
- Integrated fabricated electrodes with active sensing materials for a prototype wearable device, demonstrating proof-of-concept biomarker detection capabilities.

Python-based Data Pipeline for Material Characterization

- Automated the processing and analysis of experimental data from material characterisation equipment using Python and pandas, enabling faster iteration on material formulations.
- Developed SQL queries to extract and aggregate relevant data points from experimental logs stored in PostgreSQL, supporting trend analysis and performance evaluation.
- Built interactive Power BI dashboards to visualise key material properties and experimental results for research team review and reporting.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)